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pilssdr: A Stata command for two-sample regression estimation applying sufficient dimension reduction

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Abstract. We introduce the `pilssdr` command, which implements the plug-in least squares estimator combined with sufficient dimension reduction proposed by Hirukawa, Liu, and Prokhorov (2026). The estimator fits a linear regression when some of the regressors are missing from the dataset at hand but are available in an auxiliary dataset that contains a set of variables overlapping across both datasets. For each missing regressor, `pilssdr` models the conditional mean given the overlapping variables as a semiparametric single-index model, estimates the index coefficients by sufficient dimension reduction techniques and the link function by kernel regression, and then fits the model of interest by ordinary least squares with the imputed values in place of the missing regressors. Unlike the `msreg` command by Hirukawa, Liu, and Prokhorov (2021), which requires no more than three continuous overlapping variables, `pilssdr` attains the parametric rate of convergence for any number of matching variables. Missing regressors may be continuous or binary, and the reported standard errors account for the sampling variability of both samples as well as for the estimation error in the imputation.

Keywords: st0001, pilssdr, two-sample regression, data combination, sufficient dimension reduction, single-index model, missing regressor, kernel regression

1 Introduction

Applied researchers routinely face the situation in which the dataset at hand does not contain all the variables required by the regression model of interest. A variable may be left out of a survey because it is judged less important at the design stage, or because collecting it and hand-coding it are expensive. A textbook example is the return-to-schooling regression of Mincer (1974): unless a proxy for unobserved ability is included among the regressors, the estimated return suffers from the so-called ability bias; see Card (1995) for details. Some micro-level datasets, such as the National Longitudinal Surveys (NLS), routinely report standardized test scores that serve as ability measures,

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whereas others, such as the Current Population Survey (CPS) and the Panel Study of Income Dynamics (PSID), do not. The same problem arises elsewhere. Actual work experience, an important regressor in studies of the gender wage gap, is recorded in some surveys but not in others. Parental income, the key regressor in studies of intergenerational income mobility, is frequently unavailable in the same dataset as the offspring's income (Murtazashvili, Liu, and Prokhorov 2015). When not all relevant variables are available in a single dataset, it is common practice to integrate information from several sources, an approach known as statistical data fusion (Ridder and Moffitt 2007).

This article deals with the two-sample version of this problem. The dataset at hand, called the primary sample, contains the dependent variable and all regressors except a subset $\mathbf{x}_2$, which is missing. A second dataset, called the auxiliary sample, contains $\mathbf{x}_2$ together with a set of variables $\mathbf{x}_3$ that overlap with the primary sample. Some of the overlapping variables enter the regression model as regressors, while others are excluded from the model but are still informative about $\mathbf{x}_2$ and therefore useful for imputation. Notice that the dependent variable and $\mathbf{x}_2$ are never observed jointly, so no sample analogue of their cross moment can be formed in either sample. Consequently, two-sample instrumental-variables and two-stage least squares estimators, which build the required moments separately from the two samples (Angrist and Krueger 1992; Inoue and Solon 2010), do not apply here, and the missing regressor must instead be imputed from the auxiliary sample.

Two $\sqrt{n}$-consistent estimators are available for this setting: the matched-sample indirect inference (MSII) estimator by Hirukawa and Prokhorov (2018) and the plug-in least squares (PILS) estimator by Hirukawa, Murtazashvili, and Prokhorov (2023). MSII is implemented in Stata by the `msreg` command (Hirukawa, Liu, and Prokhorov 2021). Both estimators impute a proxy for $\mathbf{x}_2$ by regressing $\mathbf{x}_2$ on $\mathbf{x}_3$ nonparametrically in the auxiliary sample, and both therefore inherit the curse of dimensionality: the parametric rate of convergence is attained only if the number of continuous variables in $\mathbf{x}_3$ is no greater than three. Using a higher-order kernel does not resolve the problem, because the mean squared error of the imputation remains dominated by the dimension of $\mathbf{x}_3$ rather than by the order of the kernel. Unless we are willing to assume that the degree of smoothness in the conditional expectation of $\mathbf{x}_2$ given $\mathbf{x}_3$ to increase with the number of continuous components in $\mathbf{x}_3$, which is impractical. In practice, a researcher with more than three continuous matching variables must either preselect three of them, discarding information and introducing an arbitrary choice, or give up the parametric rate and the standard inference that comes with it.

Hirukawa, Liu, and Prokhorov (2026) propose a remedy for this issue. Instead of smoothing over the overlapping variables as they are, they reduce the effective dimension of the imputation problem by assuming that the conditional expectation of each missing regressor given the overlapping variables follows a semiparametric single-index model with an unspecified link function (Ichimura 1993). The index coefficients are then recovered by sufficient dimension reduction (SDR), a family of techniques that project a

high-dimensional set of covariates onto a low-dimensional subspace while preserving the information relevant for the conditional expectation. SDR is a natural fit here because in a single-index model the index coefficients are precisely the dimension reduction direction and are identified up to a scale normalization. Because each link function is a function of a scalar index, the kernel smoothing step is one dimensional no matter how many overlapping variables are used. The resulting estimator, which Hirukawa, Liu, and Prokhorov (2026) call PILS-SDR, is consistent and asymptotically normal at the parametric rate for any number of overlapping variables, and it accommodates missing regressors that are continuous as well as ones that are binary.

In this article, we introduce `pilssdr`, a new Stata command that implements the PILS-SDR estimator. Its syntax follows the familiar instrumental-variables convention: the user lists the missing regressors and the overlapping variables that enter the model inside parentheses, points the command at the auxiliary dataset with a `using` qualifier, and supplies the extra overlapping variables, or simply all of the overlapping variables, in the `extram()` option. Factor variables are allowed on either side. The SDR technique is under user control through the `sdrc()` and `sdrfv()` options, which select among sliced inverse regression (Li 1991), partial least squares (Naik and Tsai 2000), and partial inverse regression (Li, Cook, and Tsai 2007) for continuous missing regressors, and among sliced inverse regression, logit, and probit for binary ones. Reported standard errors come from a two-sample sandwich covariance estimator that accounts for the sampling variability of both samples as well as for the estimation errors in the index coefficients and in the link functions. Estimation results are posted in `e()`, so the usual postestimation commands, such as `test`, `lincom`, and `estimates table`, are available. Because `pilssdr` places no restriction on the number of continuous matching variables, it extends `msreg` (Hirukawa, Liu, and Prokhorov 2021), which is confined to at most three of them.

The rest of the article is organized as follows. Section 2 describes the two-sample setup and the single-index assumption underlying the imputation. Section 3 defines the PILS-SDR estimator, spells out the SDR estimation of the index coefficients and the kernel estimation of the link functions, and restates the consistency and asymptotic normality results of Hirukawa, Liu, and Prokhorov (2026). Section 4 documents the syntax and the options of `pilssdr`. Section 5 walks through a numerical example with simulated data. Section 6 reports a Monte Carlo study of the finite-sample performance of the estimator under the various combinations of SDR techniques. Section 7 applies `pilssdr` to a wage regression in which a test score is imputed across two datasets and compares the results with ordinary least squares and with `msreg`.

2 Model setup

Suppose that using individual-level survey data, we want to estimate the following linear regression model. For notational simplicity, we drop the subscript i, which indicates the

i-th observation.

$$y = \mathbf{x}'\boldsymbol{\beta} + u = \beta_0 + \mathbf{x}_1'\boldsymbol{\beta}_1 + \mathbf{x}_2'\boldsymbol{\beta}_2 + \mathbf{x}_{3I}'\boldsymbol{\beta}_3 + u \tag{1}$$

where $\mathbf{x}_1 \in \mathbb{R}^{d_1}$, $\mathbf{x}_2 \in \mathbb{R}^{d_2}$, $\mathbf{x}_{3I} \in \mathbb{R}^{d_{3I}}$, $\mathbf{x} = (1, \mathbf{x}_1', \mathbf{x}_2', \mathbf{x}_{3I}')'$, $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\beta}_1', \boldsymbol{\beta}_2', \boldsymbol{\beta}_3')'$, and $d := d_1 + d_2 + d_{3I}$. If all the variables in $\mathbf{x}$ are available in one dataset, we can estimate the model (1) using ordinary least squares under standard assumptions. However, in our setting, the data at hand does not contain $\mathbf{x}_2$ due to survey design or cost of data collection. Instead, we use another dataset to construct a proxy for $\mathbf{x}_2$ that can be used in the linear regression estimation.

To be more explicit, we assume there are two datasets: the primary dataset and the auxiliary dataset.

- **Primary sample A**: $\mathcal{S}_A = \{y_i, \mathbf{x}_{1,i}', \mathbf{x}_{3,i}'\}_{i=1}^n$. Notice that $\mathbf{x}_2$ is missing. $\mathbf{x}_3$ is further partitioned as $\mathbf{x}_3 = (\mathbf{x}_{3I}', \mathbf{x}_{3E}')'$, where $\mathbf{x}_{3E} \in \mathbb{R}^{d_{3E}}$. That is to say, the primary data contains some extra variables $\mathbf{x}_{3E}$ not included in the model (1) but correlated with the missing variables $\mathbf{x}_2$.
- **Auxiliary sample B**: $\mathcal{S}_B = \{\mathbf{x}_{2,j}', \mathbf{x}_{3,j}'\}_{j=1}^m$.

Thus, the variables $\mathbf{x}_3$ are observed in both $\mathcal{S}_A$ and $\mathcal{S}_B$, and they can be used to construct a proxy of $\mathbf{x}_2$. The dimension of $\mathbf{x}_3$ is $d_3 = d_{3I} + d_{3E}$. While d_3 must be strictly positive, either d_{3I} or d_{3E} is allowed to be zero.

The overlapping variables also admit a second partition $\mathbf{x}_3 = (\mathbf{x}_{3C}', \mathbf{x}_{3D}')'$ into a continuous component $\mathbf{x}_{3C} \in \mathbb{R}^{d_{3C}}$ and a discrete component $\mathbf{x}_{3D} \in \mathbb{R}^{d_{3D}}$, so that $d_3 = d_{3C} + d_{3D}$. It is the number of continuous overlapping variables, rather than d_3 itself, that governs the rate of convergence. To consistently estimate $\boldsymbol{\beta}$ using the two datasets at the parametric rate, Hirukawa and Prokhorov (2018) and Hirukawa, Murtazashvili, and Prokhorov (2023) restrict the number of continuous overlapping variables d_{3C} to be no greater than three; the number of discrete overlapping variables is unrestricted. In contrast, Hirukawa, Liu, and Prokhorov (2026) relax this restriction, and d_{3C} can be any positive number. The `msreg` command by Hirukawa, Liu, and Prokhorov (2021) implements the methods in Hirukawa and Prokhorov (2018), while the `pilssdr` command in this paper implements the methods by Hirukawa, Liu, and Prokhorov (2026). Thus, `pilssdr` is an extension of `msreg` because it allows more than three continuous overlapping variables.

In what follows, we allow two samples to be drawn from heterogeneous populations. To impute $\mathbf{x}_2$, we assume that the conditional expectation $\mathbb{E}(\mathbf{x}_{2l}^s|\mathbf{x}_3^s)$ for $l = 1, \ldots, d_2$ and $s \in \{A, B\}$ can be expressed as a semiparametric single-index model (SIM). In particular,

$$\mathbb{E}(\mathbf{x}_{2l}^s|\mathbf{x}_3^s) = g_{2l}^s(z_l^s) = g_{2l}^s(\mathbf{x}_3'\boldsymbol{\theta}_l^s) \tag{2}$$

where $g^s_{2l}(\cdot)$ is an unknown link function, and $z^s_l = \mathbf{x}^{s\prime}_3 \boldsymbol{\theta}^s_l \in \mathbb{R}$ is the linear index with the coefficient vector $\boldsymbol{\theta}^s_l \in \mathbb{R}^{d_3}$. Let $\boldsymbol{\Theta} \in \mathbb{R}^{d_3 \times d_2}$ be the index coefficient matrix and $\boldsymbol{\Theta} = [\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_{d_2}]$, and define $\mathbf{z}^s = (z^s_1, \ldots, z^s_{d_2})'$ and $g^s_2(\mathbf{z}^s) = (g^s_{21}(z^s_1), \ldots, g^s_{2d_2}(z^s_{d_2}))'$.

3 PILS-SDR estimator

The estimation procedure of PILS-SDR by Hirukawa, Liu, and Prokhorov (2026) consists of three steps. Steps 1 and 2 estimate the conditional expectation $\mathbb{E}[\mathbf{x}^A_2 | \mathbf{x}^A_3]$ by using both the primary data $\mathcal{S}_A$ and auxiliary data $\mathcal{S}_B$. Step 3 plugs the estimate of $\mathbb{E}[\mathbf{x}^A_2 \mid \mathbf{x}^A_3]$ into the main linear regression model (1) and runs the ordinary least squares (OLS) using the primary data $\mathcal{S}_A$. One key assumption for consistent estimation of the conditional expectation is that both the primary data and auxiliary data share the same index coefficient matrix and the link function. That is, we assume that $\boldsymbol{\Theta}^A = \boldsymbol{\Theta}^B = \boldsymbol{\Theta}$ and $g^A_2(\cdot) = g^B_2(\cdot) = g_2(\cdot)$. For more discussion on the assumptions of PILS-SDR estimator, see Section 3.1 of Hirukawa, Liu, and Prokhorov (2026).

The PILS-SDR estimator can be summarized in Algorithm 1. To focus on the big picture, Algorithm 1 intentionally omits the details of the index coefficient estimation and the kernel estimation of the link function. For details of the index coefficient estimation, see Section 3.1. For details of the kernel estimation, see Section 3.2.

Algorithm 1: PILS-SDR estimator

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1. **Index coefficient estimation**

   a. **SDR estimation of θ in sample B**
   In sample B, use SDR to estimate the single index for each $\mathbf{x}^B_2$ using $\mathbf{x}^B_3$. Let z_l be the index for x_{2l}. Then, by definition

   $$z_l = \mathbf{x}'_3 \boldsymbol{\theta}_l \tag{3}$$

   Denote the estimator of the index coefficient as

   $$\widehat{\boldsymbol{\Theta}}' = [\widehat{\boldsymbol{\theta}}_1, \ldots, \widehat{\boldsymbol{\theta}}_{d_2}] \tag{4}$$

   b. **predict z in samples A and B**

   $$\widehat{z}^A_l = \mathbf{x}^{A\prime}_3 \widehat{\boldsymbol{\theta}}_l \tag{5}$$

   $$\widehat{z}^B_l = \mathbf{x}^{B\prime}_3 \widehat{\boldsymbol{\theta}}_l \tag{6}$$

   for $l = 1 \ldots d_2$.

2. **Kernel estimation for the link function**

   a. **Kernel regression in Sample B**

$$\widehat{g}_{2l}(z) = \frac{\widehat{r}_l(z)}{\widehat{p}_l(z)} = \frac{\frac{1}{mh_l}\sum_{j=1}^{m} x_{2l,j}^{B} K\left(\frac{\widehat{z}_{l,j}^{B}-z}{h_l}\right)}{\frac{1}{mh_l}\sum_{j=1}^{m} K\left(\frac{\widehat{z}_{l,j}^{B}-z}{h_l}\right)} \tag{7}$$

   b. **Predict link function in sample A**
   Evaluate $\widehat{g}_{2l}(\cdot)$ at the sample-A indices to obtain $\widehat{g}_{2l}(\widehat{z}_{l,i}^{A})$ for each $l = 1, \ldots, d_2$ and $i = 1, \ldots, n$.

3. **Run the OLS using the proxy of $\mathbf{x}_2$ in sample A**

$$\texttt{regress}\ \ y \quad \text{on} \quad 1,\ \mathbf{x}_1,\ \widehat{g}_2(\widehat{\mathbf{z}}^{A}),\ \mathbf{x}_{3I} \tag{8}$$

3.1 Index coefficient estimation

There are generally three steps in the index coefficient estimation.

1. Reorder $\mathbf{x}_3$ so that one of its continuous components appear first. Together with the scale normalization in step 3, this ensures that the index coefficients are uniquely identified.
2. Estimate the unnormalized index coefficients $(\widetilde{\boldsymbol{\theta}}_1, \widetilde{\boldsymbol{\theta}}_2, \ldots, \widetilde{\boldsymbol{\theta}}_{d_2})$ using one of SDR techniques listed below.
3. Normalize each $\widetilde{\boldsymbol{\theta}}_l$ for $l = 1, \ldots, d_2$ such that its first element equals one. Denote $\widehat{\boldsymbol{\Theta}}$ as the normalized index coefficient matrix.

We can classify the SDR estimation of the index coefficient based on whether x_{2l} is continuous or binary.

1. If x_{2l} is continuous, there are three SDR methods.
   - Sliced inverse regression (SIR) by Li (1991).
   - Partial least squares (PLS) by Naik and Tsai (2000)
   - Partial inverse regression (PIR) by Li, Cook, and Tsai (2007)
2. If x_{2l} is binary, there are two methods.
   - SIR for 2 slices
   - Maximum likelihood (ML) estimation for Probit and Logit models

Sliced inverse regression

Algorithm 2 documents the implementation details of the SIR estimation.

Algorithm 2: Sliced inverse regression

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1. Partition the auxiliary sample B into H slices

   a. Sort the sample $\{x_{2l,j}, \mathbf{x}'_{3,j}\}_{j=1}^{m}$ based on the value of x_{2l}.

   b. Partition sample B into H slices with (roughly) equal size. Denote each slice as $\mathcal{I}_k$ for $k = 1, \ldots, H$. By default, $H = 10$.

2. Compute the sample mean of $\mathbf{x}_3$ in each slice.

$$\overline{\mathbf{x}}^B_{3(k)} = \sum_{j \in \mathcal{I}_k} \frac{1}{m_k} \mathbf{x}_{3,j} \tag{9}$$

   where m_k is the sample size in the slice $\mathcal{I}_k$.

3. Calculate the weighted covariance matrix of $\mathbf{x}_3$ in H slices

$$\widehat{\Sigma}_H = \frac{1}{m} \sum_{k=1}^{H} m_k (\overline{\mathbf{x}}^B_{3(k)} - \overline{\mathbf{x}}^B_3)(\overline{\mathbf{x}}^B_{3(k)} - \overline{\mathbf{x}}^B_3)' \tag{10}$$

   where $\overline{\mathbf{x}}^B_3 = \frac{1}{m} \sum_{j=1}^{m} \mathbf{x}_{3,j}$.

4. Calculate the covariance matrix of $\mathbf{x}_3$ in the full sample.

$$\widehat{\Sigma} = \frac{1}{m} \sum_{j=1}^{m} (\mathbf{x}_{3,j} - \overline{\mathbf{x}}^B_3)(\mathbf{x}_{3,j} - \overline{\mathbf{x}}^B_3)' \tag{11}$$

5. Define $\widetilde{\boldsymbol{\theta}}_l^{SIR}$ to be the principal eigenvector of $\widehat{\Sigma}_H$ relative to $\widehat{\Sigma}$. That is to compute

$$\widehat{\Sigma}_H \mathbf{v} = \lambda \widehat{\Sigma} \mathbf{v} \tag{12}$$

   and $\widetilde{\boldsymbol{\theta}}_l^{SIR}$ is the $\mathbf{v}$ associated with the largest λ.

Partial least squares

The PLS estimator is defined as

$$\widetilde{\boldsymbol{\theta}}_l^{PLS} = \widehat{\boldsymbol{\Xi}}_l \left(\widehat{\boldsymbol{\Xi}}_l' \widehat{\Sigma} \widehat{\boldsymbol{\Xi}}_l \right)^{-1} \widehat{\boldsymbol{\Xi}}_l' \widehat{\boldsymbol{\Gamma}}_l \tag{13}$$

where

$$\widehat{\boldsymbol{\Gamma}}_l = \frac{1}{m} \sum_{j=1}^{m} (\mathbf{x}_{3,j}^B - \overline{\mathbf{x}}_3^B)(x_{2l,j}^B - \overline{x}_{2l}^B) \tag{14}$$

$$\widehat{\boldsymbol{\Xi}}_l = \widehat{\boldsymbol{\Xi}}_{l(q)} = \left[\widehat{\boldsymbol{\Gamma}}_l \quad \widehat{\Sigma}\widehat{\boldsymbol{\Gamma}}_l \quad \dots \quad \widehat{\Sigma}^{q-1}\widehat{\boldsymbol{\Gamma}}_l \right]_{d_3 \times q} \tag{15}$$

$$\overline{x}_{2l}^B = \frac{1}{m} \sum_{j=1}^{m} x_{2l,j}^B \tag{16}$$

Notice that $\widehat{\boldsymbol{\Gamma}}_l$ is a $d_3 \times 1$ vector. To choose q, we follow the threshold approach of Li, Cook, and Tsai (2007). See details in Algorithm 3.

Algorithm 3: Choice of q

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1. Compute the $\widehat{\boldsymbol{\Xi}}$ when $q = d_3$.

$$\widehat{\boldsymbol{\Xi}}_{l(d_3)} = \left[\widehat{\boldsymbol{\Gamma}}_l \quad \widehat{\Sigma}\widehat{\boldsymbol{\Gamma}}_l \quad \dots \quad \widehat{\Sigma}^{d_3-1}\widehat{\boldsymbol{\Gamma}}_l \right]_{d_3 \times d_3} \tag{17}$$

2. Let $\lambda_1, \dots, \lambda_{d_3}$ be the eigenvalues of $\widehat{\boldsymbol{\Xi}}_{l(d_3)} \widehat{\boldsymbol{\Xi}}_{l(d_3)}'$ in descending order.

3. q is defined as

$$q = \sum_{p=1}^{d_3-1} \mathbb{I} \left\{ \frac{\lambda_p}{\lambda_{p+1}} > a \right\} \tag{18}$$

with $a = 1.5$.

Partial inverse regression

PIR is a hybrid of SIR and PLS. It is defined as

$$\widetilde{\boldsymbol{\theta}}_l^{PIR} = \widetilde{\boldsymbol{\Xi}}_l \left(\widetilde{\boldsymbol{\Xi}}_l' \widehat{\Sigma} \widetilde{\boldsymbol{\Xi}}_l \right)^{-1} \widetilde{\boldsymbol{\Xi}}_l' \widetilde{\boldsymbol{\theta}}_l^{SIR} \tag{19}$$

where

$$\widetilde{\boldsymbol{\Xi}}_l = \widetilde{\boldsymbol{\Xi}}_{l(q)} = \left[\widehat{\Sigma} \widetilde{\boldsymbol{\theta}}_l^{SIR} \quad \widehat{\Sigma}^2 \widetilde{\boldsymbol{\theta}}_l^{SIR} \quad \ldots \quad \widehat{\Sigma}^q \widetilde{\boldsymbol{\theta}}_l^{SIR} \right]_{d_3 \times q} \tag{20}$$

The choice of q is similar as in PLS estimator but just applied to $\widetilde{\boldsymbol{\Xi}}_{l(d_3)}$.

SDR for a binary variable

SIR for a binary variable is a special case of the general SIR technique. It differs from the regular SIR in the data partition step. For a binary variable x_{2l}, the data $(x_{2l}, \mathbf{x}_3')'$ is partitioned into two slices based on the values of x_{2l}. Thus, one slice contains data when x_{2l} equals one, and the other slice contains data when x_{2l} equals 0. These two slices need not have the same size because the values of x_{2l} may not be evenly distributed. Once the data is partitioned, the other steps of SIR are exactly the same and see details in Algorithm 2.

It is worth noting that the Probit and Logit coefficient estimators are consistent up to scale under the linearity condition, even if the error distribution is misspecified. For more discussion, see Section 2.3 in Hirukawa, Liu, and Prokhorov (2026).

3.2 Kernel estimation for the link function

For the link function estimation, we use the twiced Gaussian kernel defined as

$$K_{TG}(u) = \frac{2\exp(-u^2/2)}{\sqrt{2\pi}} - \frac{\exp(-u^2/4)}{\sqrt{4\pi}} \tag{21}$$

The theory in Hirukawa, Liu, and Prokhorov (2026) requires a kernel of order greater than two; K_{TG} is a fourth-order kernel constructed from the Gaussian kernel.

In practice, a simple rule-of-thumb bandwidth is used.

$$\widehat{h}_l = \frac{1}{2} \widehat{\sigma}_{\widehat{z}_l^B} \left(\frac{\log(m)}{m} \right)^{\overline{\alpha}} \tag{22}$$

where $\widehat{\sigma}_{\widehat{z}_l^B}$ is the sample standard deviation of $\widehat{z}_l^B$, and $\overline{\alpha} = 3/20 \in (1/8, 1/6)$.

3.3 Asymptotic properties of the PILS-SDR estimator

Hirukawa, Liu, and Prokhorov (2026) prove the consistency and asymptotic normality of

the PILS-SDR estimator. For reference, here we restate the theorems on the consistency and the asymptotic normality of the PILS-SDR estimator in the notation of this article.

Theorem 1. *Under the regularity conditions given in Section 3.1 of Hirukawa, Liu, and Prokhorov (2026),* $\widehat{\boldsymbol{\beta}}_{PS} \xrightarrow{p} \boldsymbol{\beta}$ *as* $n, m \to \infty$ *so that* $n/m \to \kappa \in (0, \infty)$.

Theorem 2. *Under the regularity conditions given in Section 3.1 of Hirukawa, Liu, and Prokhorov (2026),*

$$\sqrt{n}\left(\widehat{\boldsymbol{\beta}}_{PS} - \boldsymbol{\beta}\right) \xrightarrow{d} N\left(\mathbf{0}_{(d+1)\times 1}, \mathbf{V}_{PS}\right) := N\left(\mathbf{0}_{(d+1)\times 1}, \boldsymbol{\Phi}_{PS}^{-1}\boldsymbol{\Omega}_{PS}\boldsymbol{\Phi}_{PS}^{-1}\right)$$

as $n, m \to \infty$ *so that* $n/m \to \kappa \in (0, \infty)$, *where* $\boldsymbol{\Phi}_{PS} := \mathbb{E}(\mathbf{x}_g^A \mathbf{x}_g^{A\prime})$,

$$\begin{aligned}
\mathbf{x}_g^A &:= (1, \mathbf{x}_1^{A\prime}, g_2(\mathbf{z}^A)', \mathbf{x}_{3I}^{A\prime})', \\
\boldsymbol{\Omega}_{PS} &:= \boldsymbol{\Omega}_{PS,1} + \kappa\boldsymbol{\Omega}_{PS,2} := \mathbb{E}(\boldsymbol{\psi}_{1i}\boldsymbol{\psi}_{1i}') + \kappa\mathbb{E}(\boldsymbol{\psi}_{2j}\boldsymbol{\psi}_{2j}'), \\
\boldsymbol{\psi}_{1i} &:= \mathbf{x}_{g,i}^A \epsilon_i^A = \mathbf{x}_{g,i}^A (u_i^A + \boldsymbol{\eta}_{2i}^{A\prime}\boldsymbol{\beta}_2), \\
\boldsymbol{\psi}_{2j} &:= \boldsymbol{\psi}_{21j} + \boldsymbol{\psi}_{22j} \\
&:= \sum_{l=1}^{d_2} \mathbf{G}_l^A(z_{l,j}^B)\eta_{2l,j}^B\beta_{2l} + \sum_{l=1}^{d_2} \mathbb{E}\left[\mathbf{x}_g^A \left\{\mathbf{x}_3^A - g_{3l}^B(z_l^A)\right\}' g_{2l}^{(1)}(z_l^A)\right] \boldsymbol{\varphi}_{l,j}^B(\boldsymbol{\theta}_l)\beta_{2l}, \\
\mathbf{G}_l^A(z) &:= \mathbb{E}\left(\mathbf{x}_g^A \middle| z_l^A = z\right) \\
&= \left(1, \mathbb{E}\left(\mathbf{x}_1^A \middle| z_l^A = z\right)', \mathbb{E}\left\{g_2(\mathbf{z}^A) \middle| z_l^A = z\right\}', \mathbb{E}\left(\mathbf{x}_{3I}^A \middle| z_l^A = z\right)'\right)',
\end{aligned}$$

and $\boldsymbol{\varphi}_{l,j}^B(\boldsymbol{\theta}_l)$ *is the influence function of the index coefficient vector* $\boldsymbol{\theta}_l$.

Here $\boldsymbol{\eta}_2^s := \mathbf{x}_2^s - \mathbb{E}(\mathbf{x}_2^s \mid \mathbf{x}_3^s)$ is the reduced-form error of $\mathbf{x}_2^s$, $\epsilon^A := u^A + \boldsymbol{\eta}_2^{A\prime}\boldsymbol{\beta}_2$ is the composite error, $g_{3l}^s(z) := \mathbb{E}(\mathbf{x}_3^s \mid z_l^s = z)$, and $g_{2l}^{(1)}$ is the first derivative of g_{2l}.

For details on covariance estimation for the PILS-SDR estimator, see Supplement C of the Supplementary Materials of Hirukawa, Liu, and Prokhorov (2026).

4 Syntax

This section documents the syntax and options of `pilssdr`.

4.1 Syntax

`pilssdr` *depvar* [*x1*] `(`*x2* `=` *x3*`)` `using` *filename* [*if*] [*in*]
[`, extram(`*x3_extra*`) sdrc(`*sdrc_spec*`) sdrfv(`*sdrfv_spec*`)`]

where

- *depvar* is a *varname*.
- *x1*, *x2*, *x3*, and *x3_extra* are *varlist*s. Factor variables are allowed.
- The primary sample must be loaded in memory, and the auxiliary sample is specified by *filename*.

4.2 Options

`extram(`*x3_extra*`)` specifies the overlapping variables that are not included in the main regression model, that is, $\mathbf{x}_{3E}$. You may list only these extra variables, or you may list all the overlapping variables $\mathbf{x}_3$; any variable that already appears as *x3* in the equation is ignored, so both forms specify the same $\mathbf{x}_{3E}$.

`sdrc(`*sdrc_spec*`)` specifies the SDR technique for the continuous variables in $\mathbf{x}_2$. *sdrc_spec* is one of `sir`, `pls`, or `pir`. The default is `sir`.

`sdrfv(`*sdrfv_spec*`)` specifies the SDR technique for the factor variables in $\mathbf{x}_2$. *sdrfv_spec* is one of `sir`, `logit`, or `probit`. The default is `sir`. If a factor variable in $\mathbf{x}_2$ is not binary, `logit` and `probit` are not allowed.

5 Numerical example

We illustrate the use of `pilssdr` through a numerical example. Suppose we want to fit the following regression model:

$$y = \beta_0 + \texttt{x1}\beta_{11} + \texttt{x2_c1}\beta_{21} + \texttt{x2_fv1}\beta_{22} + \texttt{x3_c1}\beta_{31} + \texttt{x3_fv1}\beta_{32} + u$$

The variables `x2_c1` and `x2_fv1` are missing from the dataset `sample_main.dta`. Instead, we use the overlapping variables `x3*`, which appear in both `sample_main.dta` and `sample_aux.dta`, to construct a proxy for $\mathbf{x}_2$.

First, we can list the variables in `sample_main.dta`:

```
. use sample_main, clear
. describe
Contains data from sample_main.dta
 Observations:         1,000
    Variables:             9                  10 Nov 2026 11:30
-------------------------------------------------------------------------------
Variable      Storage   Display    Value
    name         type    format    label      Variable label
-------------------------------------------------------------------------------
x3_c1           double  %10.0g
x3_c2           double  %10.0g
x3_c3           double  %10.0g
x3_c4           double  %10.0g
x3_fv1          byte    %8.0g
```

```
x3_fv2          byte    %8.0g
x3_fv3          byte    %8.0g
x1              double  %10.0g
y               double  %10.0g
------------------------------------------------------------------------------
Sorted by:
```

We notice that the primary sample (`sample_main.dta`) does not contain `x2*` variables. In addition, there are four continuous variables `x3_c*` and three binary variables `x3_fv*`. These variables also exist in `sample_aux.dta` and are used as the overlapping variables. Notice that only `x3_c1` and `x3_fv1` are included in the regression model, and the other variables in `x3*` are only used as overlapping variables. Finally, the outcome variable `y` and the other independent variable `x1` are also present in this dataset.

Now, we can list the variables in `sample_aux.dta`:

```
. use sample_aux, clear
. describe
Contains data from sample_aux.dta
 Observations:         1,500
    Variables:             9                  10 Nov 2026 11:30
------------------------------------------------------------------------------
Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------------------------
x3_c1           double  %10.0g
x3_c2           double  %10.0g
x3_c3           double  %10.0g
x3_c4           double  %10.0g
x3_fv1          byte    %8.0g
x3_fv2          byte    %8.0g
x3_fv3          byte    %8.0g
x2_c1           double  %10.0g
x2_fv1          byte    %10.0g
------------------------------------------------------------------------------
Sorted by:
```

We see that both `x2*` and `x3*` are present. Thus, we can use `sample_aux.dta` to estimate the index coefficients and the link functions to create the proxy variables of `x2*` in the main sample. Now, we can run the `pilssdr` command.

```
. use sample_main, clear
.
. pilssdr y x1 (x2_c1 i.x2_fv1 = x3_c1 i.x3_fv1)  ///
>         using sample_aux,                       ///
>         extram(x3_c* i.(x3_fv*))                ///
>         sdrc(pir) sdrfv(sir)
Two sample regression with SDR        Number of obs in sample 1   =      1,000
                                      Number of obs in sample 2   =      1,500
Binary SDR:     SIR                   Number of matched vars.     =          2
Continuous SDR: PIR                   Number of overlapping vars. =          7
------------------------------------------------------------------------------
             |
```

	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
x1	1.124632	.047414	23.72	0.000	1.031702	1.217562
x2_c1	.981862	.0678703	14.47	0.000	.8488387	1.114885
1.x2_fv1	.7875508	.3890367	2.02	0.043	.025053	1.550049
x3_c1	.950987	.0990388	9.60	0.000	.7568744	1.1451
1.x3_fv1	.7665793	.1481343	5.17	0.000	.4762415	1.056917
_cons	1.140633	.14908	7.65	0.000	.8484416	1.432824

We make several remarks on the use and output of `pilssdr`.

- In the main syntax, we specify `(x2_c1 i.x2_fv1 = x3_c1 i.x3_fv1)`. We specify the missing `x2*` variables in the main sample and also the overlapping variables `x3_c1` and `x3_fv1` that are included in the model. Notice that there are other overlapping variables excluded from the model. We do not need to specify the excluded variables one by one. Instead, we can specify all the overlapping variables in the option `extram()`, and `pilssdr` will figure out which variables in `x3*` are included in the model and which ones are excluded from the model.
- We specify option `sdrc(pir)` to use the PIR estimator to estimate the index coefficients for the continuous `x2_c1`. The output header `Continuous SDR` confirms this choice.
- We specify option `sdrfv(sir)` to use the SIR estimator to estimate the index coefficients for the binary `x2_fv1`. As above, the output header `Binary SDR` shows this specification.
- The output header shows there are two matched variables and seven overlapping variables.
- The results in the output table can be interpreted as in a regular OLS regression.

6 Simulation

We conduct a simulation to study the finite-sample performance of the PILS-SDR estimator with different combinations of SDR techniques. Consider the following regression model with five regressors.

$$y^A = \beta_0 + x_1^A \beta_1 + \mathbf{x}_2^{A\prime} \boldsymbol{\beta}_2 + \mathbf{x}_{3I}^{A\prime} \boldsymbol{\beta}_3 + u \tag{23}$$

where $x_1 \in \mathbb{R}$, $\mathbf{x}_2 = (x_{2C}, x_{2D})'$, $\mathbf{x}_{3I} = (x_{3C1}, x_{3D1})'$. The overlapping variables $\mathbf{x}_3$ consist of ten variables with six continuous variables and four binary variables. $\mathbf{x}_{3I}$ consists of the first continuous and binary variable in $\mathbf{x}_3$, and the other variables in $\mathbf{x}_3$ are excluded from the model but are used as the overlapping variables. The primary sample A consists of y, x_1, $\mathbf{x}_3$. Notice that $\mathbf{x}_2$ is missing from sample A. The auxiliary

sample B consists of $\mathbf{x}_2$ and $\mathbf{x}_3$. There are 2,000 observations in sample A and 1,000 observations in sample B. The number of replications is 1,000. The true values of all coefficients are set to 1.

To estimate the coefficients, we use the proposed PILS-SDR estimator with different combinations of SDR techniques. The nine combinations are (PIR, Logit), (PIR, Probit), (PIR, SIR), (PLS, Logit), (PLS, Probit), (PLS, SIR), (SIR, Logit), (SIR, Probit), and (SIR, SIR).

6.1 Data generation

The continuous variables $\mathbf{x}_{3C}^s \in \mathbb{R}^6$ for $s \in \{A, B\}$ are drawn from a truncated multivariate normal distribution $\mathbf{x}_{3C}^s \sim TMVN(0, V_C^s; \sqrt{\mathbf{x}_{3C}^{s\prime}(V_C^s)^{-1}\mathbf{x}_{3C}^s} \leq 3)$, where V_C^s is the covariance matrix with the (i,j)th element as $\rho_N^{|i-j|}$ for $i, j \in \{1, \ldots, 6\}$. In the simulation, ρ_N is set to 0.4 in both samples, so the two populations are homogeneous.

The binary variables $\mathbf{x}_{3D}^s \in \mathbb{R}^4$ for $s \in \{A, B\}$ are generated according to correlated Rademacher variables. In particular, each element $x_{3D,k}^s$ is generated as $x_{3D,k}^s = 2\mathbb{I}(z_k^\bigstar > 0)$, where $z_k^\bigstar$ is from a four-dimensional vector $\mathbf{z}^\bigstar \sim MVN(0, V_D^s)$ with covariance matrix V_D^s having the typical elements of $\rho_D^{|i-j|}$ for $i, j \in \{1, \ldots, 4\}$. In the simulation, ρ_D is set to 0.4.

x_1 is generated as $x_1 = x_{3I,1}^A + x_{3I,2}^A + \eta^A$ with $\eta^A \sim N(0, 1)$. To generate $\mathbf{x}_2$, we need to first generate the index variable $\mathbf{z}^s = (z_1^s, z_2^s)' = (\mathbf{x}_3^{s\prime}\boldsymbol{\theta}_1, \mathbf{x}_3^{s\prime}\boldsymbol{\theta}_2)'$, where $\boldsymbol{\theta}_1 = (1, 0, -1, 1, 0, -1, 0.5, 0, -0.5, 0.5)$ and $\boldsymbol{\theta}_2 = (1, 0, -1, 1, 0, -1, -0.5, 0, 0.5, -0.5)$. The continuous x_{2C}^s is generated via $x_{2C}^s = z_1^s + 9\phi(z_1^s) + \eta_{21}$ with $\eta_{21} \sim N(0, 1)$ and $\phi(\cdot)$ is the probability density function of the standard Gaussian distribution. The binary variable x_{2D}^s is generated via $x_{2D}^s = \mathbb{I}(z_2^s + \eta_{22} > 0)$ with $\eta_{22} \sim N(0, 1)$.

Finally, y is generated via equation (23) with $u \sim N(0, 1)$.

6.2 Results

We report the simulation results for β_{21} and β_{22}, which are the coefficients for the missing variables $\mathbf{x}_2$, using the following performance measures: (i) Mean (average of parameter estimates over simulations); (ii) SD (standard deviation of the parameter estimates over simulations); (iii) SE (average of the standard errors estimates over simulations); and (iv) Rej. Rate (rejection rate of the nominal 5% test). Tables 1 and 2 summarize the simulation results for β_{21} and β_{22}, respectively.

Table 1: Results for β_{21}

SDR (cont.)	SDR (factor)	Mean	SD	SE	Rej. Rate
	Logit	0.9816	0.1096	0.1018	0.015
PIR	Probit	0.9814	0.1096	0.1017	0.015
	SIR	0.9813	0.1096	0.1027	0.014
	Logit	0.9752	0.1223	0.1039	0.020
PLS	Probit	0.9750	0.1224	0.1036	0.020
	SIR	0.9748	0.1223	0.1046	0.021
	Logit	0.9764	0.1269	0.1034	0.025
SIR	Probit	0.9763	0.1269	0.1033	0.025
	SIR	0.9761	0.1269	0.1044	0.025

Table 2: Results for β_{22}

SDR (cont.)	SDR (factor)	Mean	SD	SE	Rej. Rate
	Logit	1.0432	0.3551	0.3196	0.041
PIR	Probit	1.0448	0.3556	0.3199	0.040
	SIR	1.0492	0.3567	0.3238	0.040
	Logit	1.0568	0.3740	0.3098	0.056
PLS	Probit	1.0585	0.3750	0.3095	0.056
	SIR	1.0631	0.3758	0.3133	0.053
	Logit	1.0594	0.3831	0.3096	0.047
SIR	Probit	1.0610	0.3836	0.3098	0.048
	SIR	1.0654	0.3847	0.3140	0.046

The results show that

1. The mean of the point estimates is close to the true value. It confirms the consistency of the PILS-SDR estimator.

2. The standard deviation of the parameter estimates is close to the average of the standard errors. This suggests that the covariance estimator for PILS-SDR performs well.

3. Finally, the rejection rates are close to 5%, especially for β_{22}. For β_{21}, the rejection rate is around 2%, so the test is conservative and the confidence intervals are somewhat wider than the nominal level requires. This is a finite-sample effect that diminishes as the sample size grows.

7 Empirical application

We now apply `pilssdr` to a version of the wage equation in Mincer (1974). We want to fit the following wage equation.

$$\texttt{lnwage} = \beta_0 + \beta_1\texttt{exper} + \beta_2\texttt{exper}^2 + \beta_3\texttt{kww} + \beta_4\texttt{educ} + \beta_5\texttt{feduc} + \beta_6\texttt{meduc} + \beta_7\texttt{smsa} + \beta_8\texttt{south} + \beta_9\texttt{black} + u, \tag{24}$$

where `exper` is years of experience, `educ` is years of education, `kww` is the Knowledge of the World of Work test score, `feduc` and `meduc` are years of father's and mother's education, `black`, `smsa` and `south` are binary indicators for whether an individual is black, lives in an urban area, and lives in the south, respectively.

We can estimate regression (24) using only the dataset `card.dta` from Card (1995), as in the benchmark OLS result below.

```
. use card, clear
. regress lnwage exper c.exper#c.exper                    ///
>         kww educ feduc meduc i.smsa i.south i.black
```

Source	SS	df	MS	Number of obs	=	2,191
				F(9, 2181)	=	89.96
Model	114.487495	9	12.7208327	Prob > F	=	0.0000
Residual	308.394118	2,181	.141400329	R-squared	=	0.2707
				Adj R-squared	=	0.2677
Total	422.881613	2,190	.193096627	Root MSE	=	.37603

lnwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
exper	.0786594	.0082641	9.52	0.000	.062453	.0948658
c.exper# c.exper	-.0021546	.0004049	-5.32	0.000	-.0029486	-.0013607
kww	.0055563	.0013232	4.20	0.000	.0029615	.0081511
educ	.061174	.0051257	11.93	0.000	.0511222	.0712258
feduc	-.0018021	.0030073	-0.60	0.549	-.0076996	.0040955
meduc	.0070962	.0035015	2.03	0.043	.0002295	.013963
1.smsa	.1516693	.0186575	8.13	0.000	.1150809	.1882577
1.south	-.1111491	.0177506	-6.26	0.000	-.1459589	-.0763393
1.black	-.1320785	.0254406	-5.19	0.000	-.1819688	-.0821882
_cons	4.68611	.0815838	57.44	0.000	4.52612	4.8461

```
. estimates store ols
```

The estimation result is stored as `ols`.

Now, suppose that the variable `kww` is missing from the dataset `card.dta`. In this case, we can use another dataset `wage2.dta` from Blackburn and Neumark (1992). The dataset contains six variables `educ`, `feduc`, `meduc`, `smsa`, `south`, and `black` in addition to `kww`. All six variables are used as overlapping variables to impute the missing `kww`, where `educ`, `feduc` and `meduc` are assumed to be continuous. We want to see how

the estimation result of equation (24) changes if `kww` is imputed from `wage2.dta` using `pilssdr`.

The following are the estimation results from `pilssdr`, which are stored as `pilssdr_sdr`. Because `kww` is continuous and no SDR options are specified, `pilssdr` uses its default, `sdrc(sir)`, as the output header confirms.

```
. pilssdr lnwage exper c.exper#c.exper                    ///
>         (kww = educ feduc meduc i.smsa i.south i.black) ///
>         using wage2
```

```
Two sample regression with SDR        Number of obs in sample 1   =      2,191
                                      Number of obs in sample 2   =        470
Binary SDR:     SIR                   Number of matched vars.     =          1
Continuous SDR: SIR                   Number of overlapping vars. =          6
```

	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
exper	.0875777	.0080762	10.84	0.000	.0717487	.1034067
c.exper# c.exper	-.002323	.0004009	-5.80	0.000	-.0031086	-.0015373
kww	.0000991	.0004644	0.21	0.831	-.0008111	.0010092
educ	.072617	.0045868	15.83	0.000	.0636271	.081607
feduc	-.0006635	.0031468	-0.21	0.833	-.006831	.0055041
meduc	.0079312	.0037069	2.14	0.032	.0006658	.0151966
1.smsa	.1595206	.0179828	8.87	0.000	.124275	.1947663
1.south	-.1127755	.0178291	-6.33	0.000	-.14772	-.0778311
1.black	-.1631068	.0247333	-6.59	0.000	-.2115832	-.1146303
_cons	4.636598	.041394	112.01	0.000	4.555468	4.717729

```
. estimates store pils_sdr
```

We can also apply `msreg` (Hirukawa, Liu, and Prokhorov 2021) with `order(3)`, and the result is stored as `msreg`.

```
. msreg lnwage exper c.exper#c.exper                      ///
>         (kww = educ feduc meduc i.smsa i.south i.black) ///
>         using wage2, order(3)
```

```
Matched sample regression             Number of obs in sample 1  =      2,191
                                      Number of obs in sample 2  =        470
                                      Number of matches          =          1
                                      Order of power-series      =          3
Estimator       : twostep             Wald chi2(9)               =     785.11
Distance metric : mahalanobis         Prob > chi2                =     0.0000
```

	VI					
	Coefficient	std. err.	z	P>|z|	[95% conf.	interval]
exper	.0840744	.0082734	10.16	0.000	.0678589	.1002898
c.exper# c.exper	-.0021476	.0004202	-5.11	0.000	-.0029713	-.001324

```
        kww |   .0020755   .0052253     0.40   0.691    -.0081658    .0123168
       educ |   .0680349   .0079751     8.53   0.000     .0524041    .0836658
      feduc |  -.0005824    .003143    -0.19   0.853    -.0067427    .0055778
      meduc |   .0070538   .0039251     1.80   0.072    -.0006393    .0147469
     1.smsa |   .1557546   .0186934     8.33   0.000     .1191161     .192393
    1.south |  -.1114039   .0178276    -6.25   0.000    -.1463453   -.0764626
    1.black |  -.1517787   .0354906    -4.28   0.000     -.221339   -.0822183
      _cons |   4.654032   .1187133    39.20   0.000     4.421359    4.886706
------------------------------------------------------------------------------
Common variables : educ feduc meduc 0b.smsa 1.smsa 0b.south 1.south 0b.black
                   1.black
Matched variables: kww
Note: VI Std. Err. assumes that sample size ratio between sample 1 and sample
    2 converges to a nonzero constant and there are at most three continuous
    common variables.

. estimates store msreg
```

Finally, we compare the three estimation results from `ols`, `pilssdr_sdr`, and `msreg`.

```
. estimates table ols pils_sdr msreg, b star
```

Variable	ols	pils_sdr	msreg
exper	.0786594***	.08757774***	.08407435***
c.exper# c.exper	-.00215464***	-.00232297***	-.00214765***
kww	.00555626***	.00009908	.00207551
educ	.06117396***	.07261705***	.06803494***
feduc	-.00180207	-.00066348	-.00058241
meduc	.00709624*	.00793118*	.00705379
smsa 1	.15166928***	.15952065***	.15575459***
south 1	-.11114906***	-.11277554***	-.11140395***
black 1	-.13207849***	-.16310677***	-.15177868***
_cons	4.6861104***	4.6365984***	4.6540324***

Legend: * p<0.05; ** p<0.01; *** p<0.001

The first column shows the OLS results. The coefficients for `exper`, `kww`, and `educ` have the expected signs. The second and the third columns show the results for `pilssdr` and `msreg`, respectively. The signs and magnitudes of both the `pilssdr` and `msreg` results are similar to the `ols` results for most variables. However, for `kww`, the coefficient is insignificant in both `pilssdr_sdr` and `msreg` but significant in `ols`. For `meduc`, the coefficient is significant in `ols` and `pilssdr_sdr` but not in `msreg`.

8 Conclusion

In this article, we introduced `pilssdr`, a Stata command that fits a linear regression when some of the regressors are missing from the dataset at hand but are available in an auxiliary dataset that contains a set of variables overalpping across both datasets. The command implements the PILS-SDR estimator of Hirukawa, Liu, and Prokhorov (2026). For each missing regressor, it estimates the index coefficients on the overlapping variables by sufficient dimension reduction techniques on the auxiliary sample, estimates the corresponding link function by kernel regression using the estimated index, evaluates the fitted link at the primary sample to obtain a proxy for the missing regressor, and then fits the model of interest by OLS with the proxy in place of the missing regressor.

The practical gain over the existing `msreg` command (Hirukawa, Liu, and Prokhorov 2021) is that the dimension reduction step removes the ceiling on the number of continuous overlapping variables. Because the kernel smoothing is carried out over a scalar index rather than over the overlapping variables themselves, the estimator retains the parametric rate of convergence however many overlapping variables are supplied, and the researcher is no longer forced to select three of them and discard the rest. Missing regressors may be continuous or binary; the SDR technique is under user control through the `sdrc()` and `sdrfv()` options; and the reported standard errors come from a two-sample sandwich covariance estimator that accounts for the sampling variability of both samples and for the estimation error in the index coefficients and in the link functions.

One point deserves attention in applied work: the estimator relies on the assumption that the index coefficients and the link functions are common across two samples. This structural invariance condition is what allows the auxiliary sample to be informative about the primary one, and it should be defensible on substantive grounds before the two datasets are combined. Readers interested in the underlying theory, including the regularity conditions, the proofs, and the derivation of the covariance estimator, are referred to Hirukawa, Liu, and Prokhorov (2026) and its supplementary materials.

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